



THE AGA KHAN UNIVERSITY

eCommons@AKU

Institute for Educational Development, Karachi

Institute for Educational Development

November 2012

Knower's roles of autonomy for knowledge construction : A case of International Baccalaureate (IB) Curriculum

Raheel Tajuddin Lakhani

Aga Khan University, Institute for Educational Development, Karachi

Follow this and additional works at: http://ecommons.aku.edu/pakistan_ied_pdck

Recommended Citation

Lakhani, R. T. (2012). Knower's roles of autonomy for knowledge construction : A case of International Baccalaureate (IB) Curriculum. In *Search of Relevance and Sustainability of Educational Change : An International Conference at Aga Khan University Institute for Educational Development, November 1-3, 2012*, 187-194.

Available at: http://ecommons.aku.edu/pakistan_ied_pdck/236

KNOWER'S ROLES OF AUTONOMY FOR KNOWLEDGE CONSTRUCTION: A CASE OF INTERNATIONAL BACCALAUREATE (IB) CURRICULUM

Raheel Tajuddin Lakhani, Aga Khan University

Introduction

Autonomy is broadly defined as an ability to take responsibility for leading one's own learning (Holec, 1988). The potential for autonomy is realized when the learner is equipped with the relevant skills such as critical thinking, analytical writing and decision-making through various learning experiences (Little, 2007).

National Education Policy for Pakistan highlights one of its aims as, “to develop a self-reliant individual, capable of analytical and original thinking” (Ministry of Education, 2009, p. 11). However, operational curriculum and prevalent schooling practices fail to address this aspect of quality education. In order to conceptualize spaces for empowering autonomy, empirical evidence is needed to comprehend how a curriculum which already has a critical thinking component influences the autonomy of learners.

The research paper focuses on exploring the quality of experiencing of knowers by elucidating students' enacted roles that exhibit their autonomy for knowledge construction in the context of one such curricular instance i.e. Theory of Knowledge (TOK) course of International Baccalaureate Diploma Programme (IBDP). The core component of TOK attempts to develop critical thinking through study of epistemology. It features the ways of knowledge and areas of knowledge that a knower uses while exercising a central position (Figure 1). Therefore, the study refers the research participants as knowers (IBO, 2011).



Figure 1
Diagram of Tok Model

The research question which guides this paper is:

How do the knowers perceive their roles and meaning of autonomy in the learning processes of TOK in specific and IBDP in general?

Conceptual Framework

Feature of Autonomy

Autonomy has been associated with various decentralized concepts of education such as independent learning, student-centred learning or flexible learning (Benson, 2001). Holec (1988), one of the initial proponents of autonomy, defines it as “taking charge of one’s own learning” (p. 3). Dam (1995) refines this definition by including the concept of readiness which entails both capacity and willingness to take charge to act both independently and in co-operation with others. Capacity involves the ability to perform tasks with independence, whereas willingness highlights the intrinsic motivation or strength of intention.

Characteristics of Autonomous Learner

Some common features of autonomy can be identified from the multitude of definitions that are at our disposal. These features include internal reflection, meta-cognition awareness and competence. Autonomy is practiced when one’s beliefs and desires are a result of “rational and informed reflection” (Carter, Kramer, & Steiner, 2007, p. 323). Consistent with the feature of reflection, autonomy is also about meta-cognitive awareness that is “consciousness” of one’s beliefs and actions. It consists of focus and self-evaluation of what one knows, what one is currently doing and what is one’s cognitive and affective state/capacity is (Hacker & Dunlosky, 1998; Wilkins, 1996; Wenden, 1991). This helps an individual to make informed decisions about his/her learning and to readily employ the resources which are available.

Participants in research of Chan (2001) highlighted some attributes of autonomous learners such as being highly motivated, inquisitive, well-organized, hardworking and interested in the topic being explored, active initiators, flexible and make use of every opportunity to learn (pp. 512–513). These features provide a framework to highlight autonomous practices of knowers.

Methodology

Since the study explored role of the knowers in exercising autonomy for knowledge construction in a bounded context of IBDP’s TOK course and related curricular spaces, therefore Qualitative Case Study was used as a research paradigm to explore, observe, and understand perceptions and opinions of knowers regarding their learning practices and experiences (Gerring, 2005).

The study was conducted in a private school The Urban Academy in Karachi, Pakistan which follows two-year long IBDP curriculum at higher secondary level (16-19 years). The study participants (Table 1) including three students from Year 1 of IBDP: two female (Ariana, Sabaina) and a male student (Ayaan) and the TOK teacher (Simak) were chosen using purposive sampling to select information-rich cases. The selection of participants was based on relative extents of classroom participation and adequate participation of both genders in the research.

Table 1

Knower	Age	Prior Schooling	Degree of Participation
Ariana	15	Transited from MYP. Prior to that, she has received her schooling in Kenya where she studied in a school which taught O levels curriculum.	An active participant in the classroom discussions
Ayaan	19	Joined IBDP after completing his Matriculation and getting some work experience at a software house.	An alternate way of participation. He always comes up with the questions on the readings which he discusses with the teacher in spare time.
Sabaina	16	Joined IBDP after completing her O levels (secondary education).	Relatively participates less than others
Simak	-	He has received B.A. in Literature and Economics from a foreign liberal art college. He has been teaching for last five years and completed his M.Ed. in Critical and Creative Thinking from a foreign university. He has joined The Urban Academy from the recent academic year as a part-time teacher.	Pedagogue for the TOK

The study employed multiple methods for data generation such as tape-recorded semi-structured interviews and classroom observations. Study guides, textbooks and students' essays were also reviewed as part of document analysis. The data from multiple sources was transcribed/reviewed/triangulated and analysed continually to identify emerging themes (Merriam, 1998) which are discussed below.

Findings

The data suggest the knowers enact the roles of a questioning mind or an inquirer, reflective thinker and creative communicator.

Knower as a Questioning Mind

The findings suggest that one of the core practices of knowers in learning processes is that of questioning. Students were observed asking questions during discussions, employing inquisitiveness during their independent readings and using e-tools to find information during classroom sessions. Such questions led to meaning-making and transformed their constructs of various concepts.

For instance, in one of the assignments, Ariana used an example which was not discussed in the class. When she was asked about how and why she incorporated a fallacy of knowing in that assignment which was beyond classroom learning, her response resonated the same prominence of innate interest or curiosity in the topic of study:

Because I was thinking and I started questioning what would be another fallacy in this kind of knowing. Curiosity made me read ahead, in the book and on the internet and I wanted to put those learning in the assignment. Your book cannot answer all your questions so you have to look into more sources. The main question of assignment matter but my own questions also matter equally to me. (Interview with Ariana, March 14, 2012)

Sabaina also shared similar views regarding her learning engagements where discussion plays a vital role in directing her towards independent searches, “For me, discussion leads to questioning things, then I think about it and research on it on my own.” (Interview with Sabaina, March 12, 2012)

On the other hand, the data indicates that if knower does not have self-confidence on either the ability to construct knowledge or adequacy of their own constructed ideas then they would require external support and affirmation. In Sabaina’s situation, she is still not sure about how to carry out learning tasks because she doubts whether she is fulfilling the task expectations. As a result, despite of knowing the content, still she is sometimes unable to translate that knowledge into an organized process of creating a knowledge product. It creates a challenge for her autonomous learning practices. According to Sabaina:

The reason I want supervision is that I guess I am not quick enough; I don’t have enough belief in myself. You get confused but when you know that it’s the way everyone is doing it then you do your work accordingly. A lot of times, in the essays, I have not been able to complete them as I didn’t know how to go about it. Sometimes when I reach the required word limit, I just submit the essay and don’t feel like working hard over it. (Interview with Sabaina, March 08, 2012)

Furthermore, in such a situation, the freedom to think loses its usefulness when the person is not skilled to deal with that freedom.

Knower as a Reflective Thinker

The data reports that knowers were actively engaged in constructing knowledge during classroom discussions and essays by reflecting on their daily life experiences. For example, Ayaan, in one of his essays, had reflected and transformed his conception of hedonism or self-pleasure in light of community service. The glimpses of the same process also resonate in the opinions of Ariana, “TOK gives you a certain bridge to reflect. Like does an activity appeal to emotion or reason and why is that important and how it is helping you to learn something?” (Interview with Ariana, March 08, 2012)

Knowers also shared the content in the book helps knowers to build upon their knowledge as it challenges their thoughts. Students also feel that pedagogue helps them to understand the concepts. Moreover, Simak also thinks that “TOK helps students to understand themselves in relation to the rationale of what they are doing.” (Teacher Interview, February 8, 2012). In document analysis, it was noted that every chapter in the book starts with a list of quotations and proverbs. Some of them are also part of activity boxes which are placed throughout the chapter;

sometimes images being the object to reflect upon. It is up to the knower whether they engage in these reflections. During interviews, Ariana also shared a similar opinion,

It is engaging and interesting to read because of the informal way in which it is written, like someone talking to you. It also has activities, pictures and linking questions to other topics which help me to reflect from multiple perspectives. Similarly books of other subjects also have boxes (bubbles) which tell how the content relates to TOK. (Interview with Ariana, March 14, 2012)

Knower as a Creative Communicator

Research data provided evidence regarding creative approach for presenting knowledge. In a knowledge piece on using reason in daily life, Ariana contained originality of argument. During this assignment, she placed the self practically in the centre of discourse and actually analysed minute activities that she herself undertakes from morning to night was a novel approach of understanding ‘reason’ as a way of knowing. Every creative endeavour involves some risk-taking because it drives beyond the convention. She communicated:

When I wrote an assignment through a different approach, I was not really sure how it would be received. I have always written objective pieces which have concrete or established concepts. This time I felt like taking a risk and included subjective personal experiences. (Interview with Ariana, March 14, 2012)

Similarly, for example, Sabaina’s first assignment for TOK course was filled with images and she thought these visuals helped aid the communication of her views regarding the topic of discussion. She elucidated, “It is about the writer’s choice because some people can express their ideas better through pictures. Sometimes you know, instead of writing a paragraph, you put a picture and that speaks all” (Interview with Sabaina, March 12, 2012). It can be inferred that her assignment took images from outside sources and tried to interpret those images by weaving in her personal ideas to construct a new meaning. It was a risk to be innovative as creativity comes with a price. Her assignment was not that well-received because in their school’s context, students have to follow conventions of essay-writing; they can be creative through words only and using images are not encouraged.

However, it was identified during document analysis that the study guide itself allows videos or posters to be part of presentations and there is no explicit mention of not using metaphors. Yet, the data from interview suggests that knowers have built these perceptions based on the guidelines and feedback they have received on their assignments. This has limited the creativity and originality of their ideas and therefore their autonomy for knowledge construction as well.

Discussion

In light of findings, knowers become autonomous through two processes/conditions; firstly, development of their competence by provision of knowledge and skills of critical thinking. The findings show that each knower is willing to take the responsibility of reflecting upon their daily life experiences, areas of knowledge and larger existential questions. They have started exploring depths of their thoughts and the complexities of subsequent questions that emerge out of those intellectual wanderings which are evident from their responses in interviews. The reasons for

their behaviours, choices and actions of learning come from inside within the consciousness of self (Benn, 1988). As mentioned in the findings, participants feel that IB curriculum in general and specifically TOK course encourages the questioning and justification of one's beliefs and claims of knowledge. Little (1991) considers autonomy as a capacity for critical reflection. The knowledge of epistemology helps students to think analytically in TOK and other learning endeavours. Secondly, knower's act autonomously when they have self-confidence about their capacity of critical reflection and meaning-making.

As a result, knowers are more willing to take risks, present and defend their ideas in creative ways, value their own questions and have confidence in their independent quests. Surprisingly, they recognize constraints and do not become overly affected by those limitations. Conversely, they manoeuvre their ways to work in those constraints. For example, students are not allowed to use images in their written assignments; however students still try to be creative and imaginative by quoting a variety of examples.

Nevertheless, not everyone is at peace with this task. It is not an easy task as it creates disequilibrium in the prior cognitive and current emotional state; social and academic expectations create such disturbances. Pedagogue first needs to comfort students that knowledge would certainly create paradoxes because dissonance is vital for constructivist learning and it is also the agency of knower in knowledge construction which proposes the solution (Moon, 2008). Moreover, the students should be aware that there is no one right answer; rather it is the intellectual rigor which counts the most.

The findings show that knower is agency for knowledge construction is related to autonomous interdependence (Atkins, 2009). Even when a knower interprets his/her personal experiences independently, those experiences are still rooted in a social interaction. Even one a person is engaged in an epistemological course to build improve critical thinking, s/he requires a classroom to build the competence or a book written by another person to learn those concepts. Reading is not the end of an automous learning experience. Knowers' need of spaces for interaction, discussion and dialogue in the classroom is one of the most dominant findings of this research.

Conclusion and Recommendations

The data suggest that knowledge construction starts from questioning which then leads to the search for explanations. These explanations arise by thinking upon content resulting from their searches and reflecting upon life experiences. Inferred explanations are then communicated through classroom discussions and their knowledge-based essays. The content of the course and one's confidence in critical reflection skills helps autonomous learning.

Briefly, it can be asserted that, the roles of being Inquirers, Thinkers, Risk-takers, Reflective and Creative mentioned in the learner profile and also their enactments highlight the learners' initiative, critical thinking, diverse perspectives, creative expression and self-reflection.

In light of the research findings, it is recommended that the curriculum should provide opportunities where their voices and interpretations are essential.

1. Introducing the use of visual mediums for meaning-making and representation as part of their written assessment.

2. Scaffolding through guided approach, reflective questions and proper communication of role expectations should be provided by pedagogue in independent readings.
3. Employing cooperative learning strategies such as debates, drama and case-studies to support relatedness in autonomy by provide spaces for interaction among students to share their concerns while performing independent tasks.

References

- Atkins, K. (2006). Autonomy and autonomy competencies: a practical and relational approach. *Nursing Philosophy*, 7, 205-215.
- Benn, S. I. (1988). *A theory of freedom*. Cambridge: Cambridge University Press.
- Benson, P. (2001). *Teaching and researching autonomy in language learning*. Essex: Longman.
- Carter, I., Kramer, M. H., & Steiner, H. (2007). Introduction to Part VI. In *Freedom: A philosophical anthology* (pp. 323-325). Victoria: Blackwell Publishing.
- Chan, V. (2001). Readiness for learner autonomy: What do our learners tell us? *Teaching in Higher Education*, 6(4), 505-518.
- Dam, L. (1995). *Learner Autonomy 3: From theory to classroom practice*. Dublin: Authentik.
- De Lagemaat, R. (2011). *Theory of knowledge for the IB diploma*. Cambridge: Cambridge
- Gerring, J. (2005). *Case study research*. New York: Cambridge University Press.
- Hacker, D. J., & Dunlosky, J. (1998). *Metacognition in educational theory and practice*. Mahwah: Lawrence Erlbaum.
- Holec, H. (1988). *Autonomy and self-directed learning*. Strasbourg: Council of Europe.
- IBO. (2009). *IB learner profile*. Geneva: International Baccalaureate Organization.
- IBO. (2011). *IB diploma programme curriculum*. Singapore: International Baccalaureate Organization.
- Little, D. (1991). *Learner Autonomy: Definitions, issues and problems*. Dublin: Authentic.
- Little, D. (2007). Language learner autonomy: Some fundamental considerations revisited. *Innovation in Language Learning and Teaching*, 1(1), 14-29.
- Macaro, E. (1997). *Target language, collaborative learning and autonomy*. Clevedon: Multilingual Matters Ltd.
- Merriam, S. B. (1998). *Qualitative research and case study applications in education*. San Francisco: Jossey-Bass.
- Ministry of Education. (2009). *National educational policy*. Islamabad: Government of Pakistan.
- Moon, J. (2008). *Critical tinkering: An exploration of theory and practice*. London: Routledge.
- Reinders, H. (2011). From autonomy to autonomous language learning. In A. Ahmed, G. Cane, & M. Hanzala (Ed.), *Teaching English in multilingual contexts* (pp. 37-52). Newcastle: Cambridge Scholars Publishing.

- Wenden. (1991). *Learner strategies for learner autonomy*. New York: Prentice Hall.
- Wilkins, S. (1996). Fostering independence with metacognition. In *Proceedings of_Autonomy 2000: The development of learning independence in language learning* (pp. 254-263). Bangkok: King Mongkut's Institute of Technology.